The Elbow Method

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Introduction

The elbow method is the simplest heuristic for choosing the number of clusters in k-means or any partitioning algorithm. The within-cluster sum of squares (WSS) decreases monotonically with k, but the rate of decrease slows once the natural cluster structure has been captured. Plotting WSS against k reveals a "bend" — the elbow — at the point where additional clusters bring diminishing returns. The method is unavoidably subjective but widely used because it is fast, intuitive, and works well when the data have well-separated clusters.

Prerequisites

A working understanding of k-means clustering, the within-cluster sum of squares as a quality criterion, and the relationship between k and within-cluster compactness.

Theory

For partition $C = \{C_1, \ldots, C_k\}$ of the data, $\mathrm{WSS}(k) = \sum_{j=1}^{k} \sum_{x \in C_j} \|x - \bar{x}_j\|^2$ is the total within-cluster sum of squared distances to centroids. As k grows, WSS necessarily decreases — at $k = n$ it reaches zero. A genuine cluster structure produces a sharp bend at the true number of clusters, after which adding more k subdivides homogeneous regions and yields only modest reductions. The bend is identified visually or, more reproducibly, by algorithms such as Kneedle that compute a normalised distance from each point on the curve to a chord between the endpoints.

Assumptions

A genuine cluster structure exists in the data; if no clusters are present, no clear elbow appears. Variables are standardised so distances along each axis are comparable.

R Implementation

```r
library(factoextra)

X <- scale(iris[, 1:4])

# Elbow plot
fviz_nbclust(X, kmeans, method = "wss", k.max = 10)

# Manual: compute WSS across k
wss <- sapply(1:10, function(k)
  kmeans(X, centers = k, nstart = 25)$tot.withinss)
plot(1:10, wss, type = "b",
     xlab = "Number of clusters k",
     ylab = "Within-cluster sum of squares")
```

Output & Results

A monotone decreasing WSS curve. For the standardised iris data, the bend appears at $k = 3$, consistent with the three species. `fviz_nbclust()` automates the standard elbow plot with `ggplot2` styling and works with k-means, PAM, and CLARA via the `FUNcluster` argument.

Interpretation

A reporting sentence: "The elbow plot showed a clear bend at $k = 3$ (within-cluster sum of squares dropped from 681 at $k = 1$ to 140 at $k = 3$ and only marginally further to 96 at $k = 10$); we chose $k = 3$, consistent with the silhouette method's optimum and with the known species structure." Always couple the elbow visual with at least one quantitative criterion (silhouette, gap statistic).

Practical Tips

- Pair the elbow with the silhouette method (`fviz_nbclust(..., method = "silhouette")`) and the gap statistic (`method = "gap_stat"`); concordance across all three increases confidence in the chosen k.
- If no clear elbow appears, the data may not have a natural cluster structure; do not force a k where the WSS curve is smooth.
- Standardise variables before clustering; variables with larger variance dominate WSS without standardisation and distort the elbow.
- Use `nstart = 25` or higher when computing WSS; multiple random starts avoid local minima that produce noisy WSS curves.
- The Kneedle algorithm (R package `kneedle`) automates elbow detection by finding the point of maximum curvature; useful for batch processing many datasets.
- Elbow alone is insufficient justification for a chosen k in publications; report it alongside silhouette and gap-statistic results.

R Packages Used

`factoextra` for `fviz_nbclust()` ggplot-style elbow, silhouette, and gap-statistic plots; `cluster` for the underlying clustering algorithms; `NbClust` for 30+ cluster-validation indices in one call; `kneedle` for automated elbow detection.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE